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Neogene clockwise tectonic rotation of the eastern Transverse Ranges, California, suggested by paleomagnetic vectors

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ABSTRACT

Results from a paleomagnetic study of Neogene volcanic rocks situated within and around the Pinto basin region of the eastern Transverse Ranges suggest that the ranges have undergone a clockwise tectonic rotation of ~41.4° ± 7.7° since late Miocene time. Both normal and reversed polarities are observed, and inclination data indicate no significant flattening. The rotation was accompanied by progressive sinistral slip on the several east-west–trending faults which bound and transect the eastern Transverse Ranges. A computation using the observed amount of cumulative left slip on the east-west–trending faults of the eastern Transverse Ranges (~50 km) and the spacing between the Pinto Mountain and Salton Creek faults which bound the region (~70 km) indicates that ~35° of clockwise rotation is required, which roughly agrees with the paleomagnetic data. Removal of the eastern Transverse Ranges rotation in a reconstruction shows that the distributed dextral shear across the Mojave Desert region to the north must have been on the order of 100 km since late Miocene time. From the results of this study, we infer that crustal rotation of the eastern Transverse Ranges began later and is of less magnitude than that of the western Transverse Ranges.

INTRODUCTION

The tectonic history of western California has been complicated by the network of complex structural features and displaced terranes resulting from North American–oceanic plate interactions. Beck (1976) noted the predominance of discordant, east-deflected, and/or flattened (anomalously shallow) paleomagnetic directions found along the western North American margin. He interpreted these as indicating that dextral shear between the plates has translated and/or rotated clockwise fragments of both continental and oceanic crust into place as exotic blocks or "terranes" along the continental margin. Cox (1980) has provided a summary of the various mechanisms proposed by several authors to account for the implied deformations (for example, rotation of blocks loosened by extension in pull-apart basins, rotation of conjugate shear faults, translations and rotation of continental microplates, and so on). Central to all of the mechanisms is the concept that brittle crustal blocks must be detached at depth and allowed to translate and rotate about a near-vertical axis above this detachment surface in response to shearing forces generated by more ductile material beneath. McKenzie and Jackson (1983) have shown this to be a viable mechanism by using fluid-mechanics theory to demonstrate the effect of a fluid under shear on overlying floats.

Figure 1 shows a generalized structural map of southern California outlining the location of the present study area. The area occupies ~4,400 km^2 within the eastern Transverse Ranges and contains several volumetrically small basalt fields. As in the western Transverse Ranges, the study area is characterized by several east-west–trending sinistral faults. Paleomagnetic data from the western Transverse Ranges suggest that ~90° of clockwise rotation has occurred in this region since early Miocene

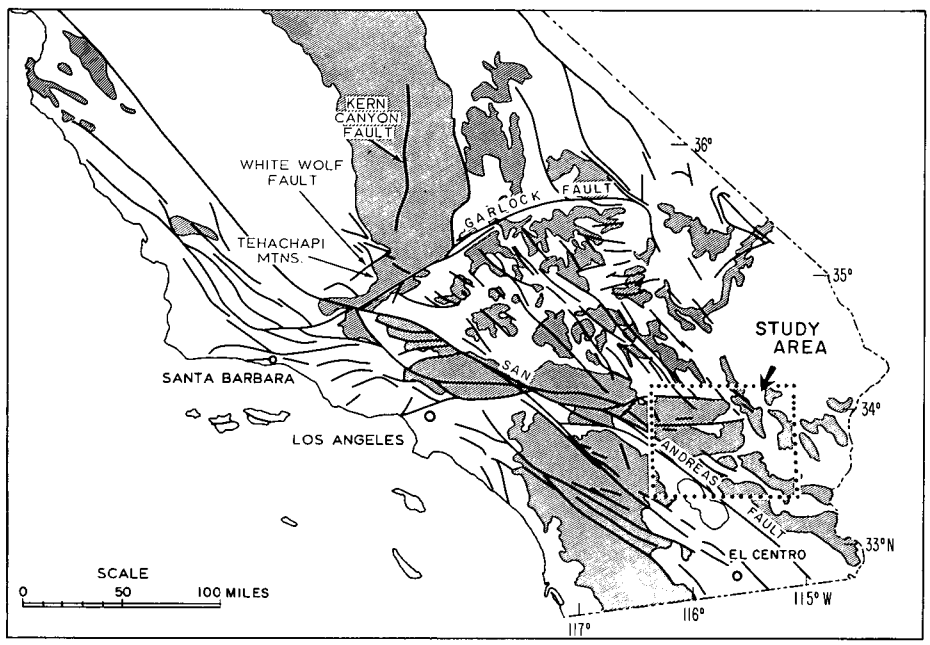


Figure 1. Generalized structural map of southern California, showing the location of the study area within the eastern Transverse Ranges region. Geology is from the U.S. Geological Survey and California Division of Mines and Geology (1966) geologic map of California, scale 1:2,500,000. Shaded area indicates crystalline basement rocks of Mesozoic and older age.

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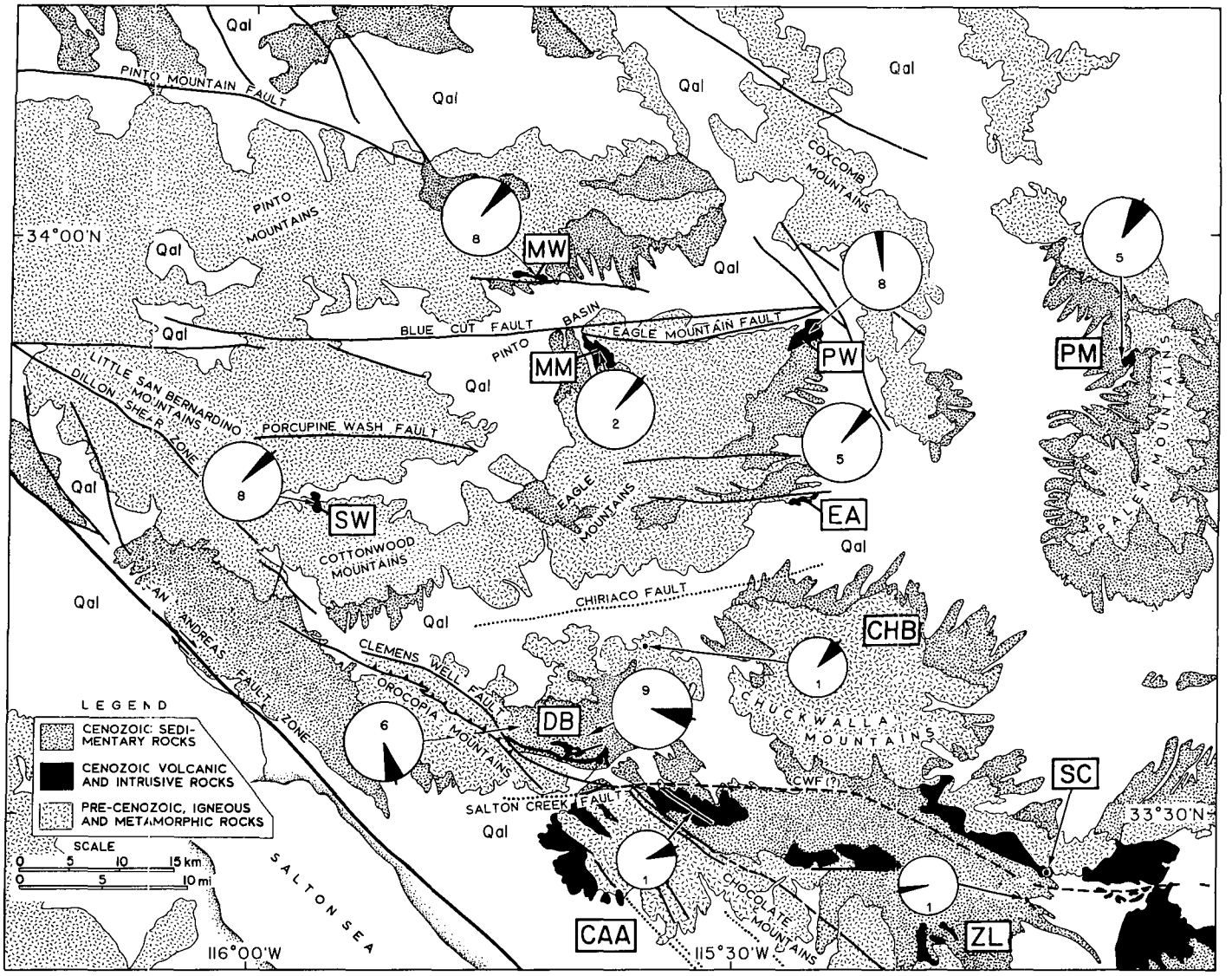


Figure 2. Generalized geologic map of the study area, showing sampling localities, the number of flows sampled, and paleomagnetic declinations. The mean declination at each locality is shown with the horizontal projection of the 95% confidence limit on the mean. Smaller circles are for localities having only one unit (flow). Locality symbols for the Pinto basin region are SW = Smoketree Wash, MW = Mission Well, MM = Mystery Mines, PW = Pinto Well, EA (from Terres, 1984), PM = Palen Mountains. Symbols for southern region are DB = Diligencia basin (Terres, 1984), ZL = Zetta Lane, SC = Southern Chuckwallas (no mean result), CAA and CHB (from Terres, 1984). Also shown is the possible eastward extension of the Clemens Well (right lateral) or Salton Creek (left lateral) faults as a dashed line (after Powell, 1981). Localities in Pinto basin region are of basalts of late Miocene age; localities in southern region are of basalts, dacites, and rhyolites of early and middle Miocene age. (Geology is from U.S. Geological Survey California state sheets, scale 1:250,000: Salton Sea sheet, compiled by C. W. Jennings, 1967; Needles sheet, compiled by C. C. Bishop, 1963; San Bernardino sheet, compiled by T. H. Rogers, 1967; and Santa Ana sheet, compiled by T. H. Rogers, 1965.)

time (Kamerling and Luyendyk, 1979; Luyendyk and others, 1980, 1985a; Terres, 1984; Hornafius, 1985). On the basis of this work and following the general ideas of Freund (1974), Dibblee (1977), and Livicarri (1979), Luyendyk and others (1980) developed a model for Neogene crustal rotation of southern California. In this model, rigid crustal blocks bounded by sinistral faults rotate clockwise as a result of progressive dextral slip on northwest-trending faults, which respond to the dextral shear between the Pacific and North American plates. Inherent in this model are that (1) the entire Transverse Ranges region has rotated 90° clockwise, the present east-west–trending faults probably originating as tension gashes within the shear couple, and (2) the regions bounded by the dextral faults do not rotate. Garfunkel (1974) has suggested that the Mojave Desert region (transected by northwest-trending, dextral faults) and eastern Transverse Ranges have deformed in response to regional east-west–directed left shear resulting from differential extension between the Basin and Range province to the north and southern California borderland province to the south. This shear regime would result in the Mojave Desert region rotating as much as 30° or 40° counterclockwise while allowing for no rotation of the eastern Transverse Ranges.

The purpose of the present study is (1) to

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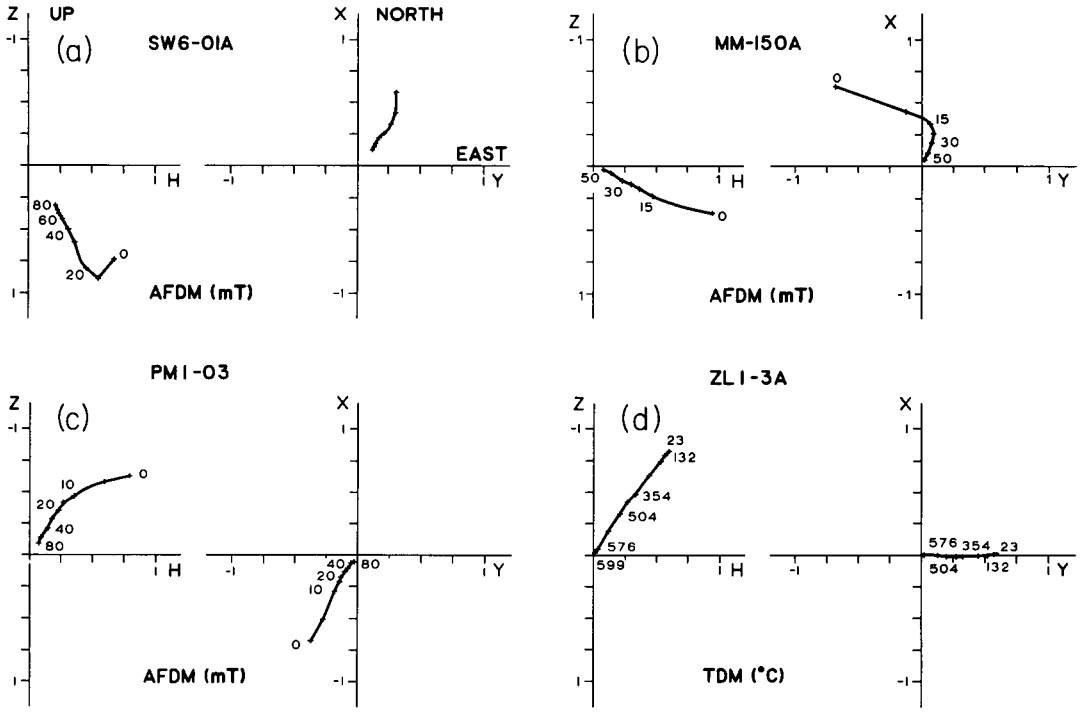


Figure 3. Normalized orthogonal vector diagrams showing typical demagnetization behavior for four representative samples. (a) Smoketree Wash sample showing removal of viscous component and stable behavior. (b) Mystery Mines sample showing lower coercivity but stable behavior and a lesser clockwise declination and anomalously shallow inclination. (c) Palen Mountains result showing stable behavior and a reversed-polarity, clockwise-deflected direction. (d) Zetta Lane thermal demagnetization result showing very stable demagnetization behavior but an anomalous declination.

constrain the eastward extent, amount, and timing of the rotations observed for the eastern Transverse Ranges and then (2) to modify the models of Garfunkel (1974) or Luyendyk and others (1980) accordingly.

PALEOMAGNETIC RESULTS

Samples were collected from five localities around the Pinto basin region (Pinto, Eagle, and Little San Bernardino Mountains), from one locality in the Palen Mountains to the east, and from four localities in the Chuckwalla, Orocopia, and Chocolate Mountains to the south (Fig. 2). Samples were taken either using a gas-powered portable drill or especially in inaccessible or restricted areas, as oriented hand samples. All sample orientations were measured using a Brunton compass; local magnetic gradients were checked for with a compass. Locations where anomalies were present were avoided. At some locations, core orientations were determined by using sun-sighting techniques. The over-all consistency of results at each locality suggests that errors attributed to hand sampling or local gradients are negligible.

All samples were subjected to step-wise alternating field (a.f.) demagnetization in fields ranging to as much as 120 milliTesla (mT; SI units according to Shive, 1986), and most (166 in total) yielded stable "characteristic" directions determined with the aid of both equal-area and orthogonal vector plots (Zijderveld, 1967; Fig. 3). In addition, representative samples from each locality were subjected to either step-wise or continuous thermal demagnetization. This was necessary for several samples exhibiting anomalously high coercivity of remanence. Directions were averaged in cases in which both a.f. and thermal demagnetization were done on separate specimens from the same core or hand sample.

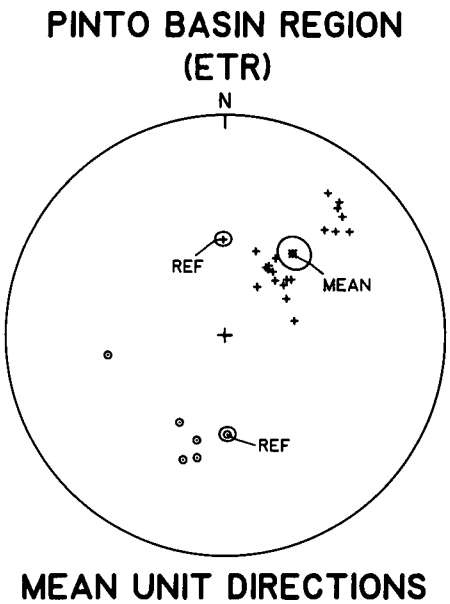


Figure 4. Individual unit means for the clockwise-deflected directions at the localities in Table 2. Mean direction and reference direction (Irving and Irving, 1982) are also shown.

Pinto Basin Region

With the exception of the Pinto Well (PW) locality, paleomagnetic declinations from all localities around Pinto basin and in the Palen Mountains are significantly clockwise-deflected from the late Miocene reference pole of Irving and Irving (1982) and present geographic north (Fig. 4). The samples were collected from small basalt fields consisting of flows typically 3–5 m thick and spanning no more than 150 m in total stratigraphic thickness. The basalts range from red to dark gray in color and from massive to highly vesicular.

The undisturbed and stratified appearance of the Mission Well, Smoketree, Washington, and EA basalt fields suggests that the flows here were probably extruded from fissures onto a flat topographic surface (no volcanic centers are observed). The flows at Mystery Mines, Pinto Well, and Palen Mountains localities, however, are more irregular in form and have variable steep dips. These were probably extruded onto a sloped, incised topography. The Palen Mountains field appears to be a portion of a shoulder of a volcanic center. Structural attitudes were estimated as accurately as possible from exposed flow tops (Table 1). No interbedded sediments were observed. Although these attitudes would seem to indicate post-volcanic deformation, the flows at these localities look structurally undisturbed. Consequently, we believe that *in situ* rather than tectonically corrected paleomagnetic directions are more accurate. Tectonic corrections steepen directions to anomalous inclina-

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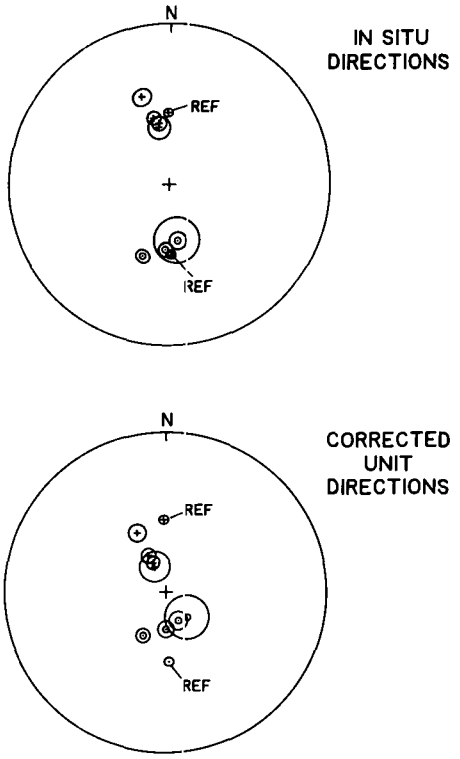


Figure 5. Equal-area projection of uncorrected and corrected unit (flow) average directions and 95% confidence limits from Pinto Well (PW in Fig. 2). Upon untilting 15° about strike, the results steepen away from the normal and reversed reference directions. Because the possibility of remagnetization is probably precluded by the presence of both normal and reversed polarities, this test most likely indicates (1) that the flow dips observed are initial dips and (2) that this locality has undergone no net rotation.

tions. At PW, the correction steepens the directions away from the reference direction (Fig. 5). There is no geologic evidence in the region that supports late Tertiary folding. Tertiary alluvial-fan and lacustrine deposits are undeformed. We conclude, therefore, that the present dips of the flows represent original attitudes.

Southern Region

To the south of the Chiriaco fault (Fig. 2), four localities were sampled within the Orocopia and Chuckwalla Mountains (SC, CAA, CHB, ZL). The only samples which exhibit stable characteristic magnetizations are from a dacite flow at the Zetta Lane (ZL) locality, a felsic dike at the CHB locality, and a basalt flow at the CAA locality (Table 2).

Results from two study areas comprising 15 sites within the Diligencia basin (DB) represent work on early Miocene flows by Terres (1984). The extreme apparent clockwise rotation here is attributed to the proximity of the dextral Clemens Well fault (CWF in Fig. 2; Terres, 1984). Other than the DB results, secular variation cannot be adequately averaged in this region due to the few units and uncertain age control. Further, it is not known whether the anomalous declinations of DB include a regional rotation effect. Little regional tectonic significance can, therefore, be attached to these data at present.

Interpretation of Pinto Basin Results

Magnetization experiments (discussed below) and orthogonal vector plots suggest that the characteristic directions selected are in fact primary directions due to thermoremanent magnetization (TRM). The possibility of remagnetization of these subaerial and basically unaltered basalts is very remote, especially considering the dual polarities observed at PW (Fig. 5) and in the region (Figs. 4 and 6), which indicate that the reversal test is passed.

The limited number of flows at each locality and the probably short time interval spanning their extrusion most likely preclude the averaging of secular variation at any one location. The abnormally high and low values of the precision parameter (kappa) at each locality, and also the anomalous inclination values, support this interpretation (Table 2). Complete averaging of the dispersion in directions due to secular variation should result in a kappa near 35 (McFadden, 1980). If the clockwise-deflected directions from all of the units (27) or all of the localities (5) in the Pinto basin are averaged, kappa is a more reasonable 20 or 24 but is still low (Table 2). This suggests partial averaging of secular variation over the 4-m.y. time span indicated by the K-Ar ages. Any differential tectonic rotation between the localities, however, would also contribute to the dispersion.

Of the 27 clockwise-deflected directions, five are reversely magnetized (Fig. 4). Compared to the Miocene reference direction for the stable North American craton (Irving and Irving, 1982), the average direction for the 27 units suggests a clockwise rotation of 41.4° ± 7.7° and an inclination flattening of 2.1° ± 5.4° (Table 2).

Pinto Well (PW) is the only locality which does not show a clockwise declination anomaly. Because remagnetization here is unlikely, it is assumed that this locality has undergone no net rotation. It may be argued that because the PW basalts are the youngest studied (7.8 Ma; Table 2), the rotation was completed before they were laid down. The EA basalts at 10.2 Ma appear rotated, which brackets the rotation

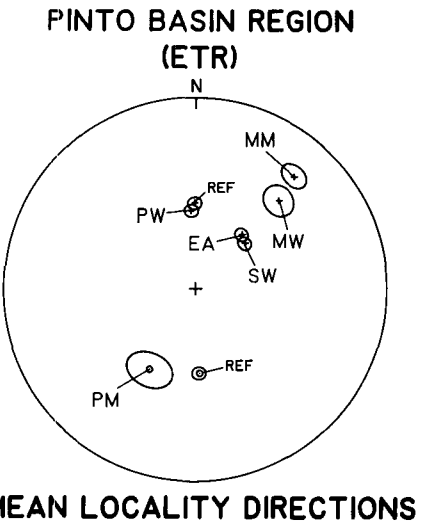


Figure 6. Equal-area projection of paleomagnetic results from the Pinto basin region, eastern Transverse Ranges. Mean paleomagnetic directions and corresponding limits of 95% confidence are shown; crosses indicate normal polarity, and open circles indicate reversed polarity. Locality symbols are as in Figure 2. Miocene reference directions are from Irving and Irving (1982). A reversal test is passed, the reversed Palen Mountains result being as equally clockwise deflected as are normal polarity results (within confidence limits). PW is the only apparently unrotated locality; reversed directions here were projected through the origin to plot with the normal directions recorded at this locality.

TABLE 1. LOCATIONS AND STRUCTURAL ATTITUDES

Site	Latitude (degrees)	Longitude (degrees)	Attitude, strike/dip
SW	33.78N	115.93W	073/6NW
EA	33.78	115.42	090/5S
MW	33.97	115.72	171/9E
PM	33.88	115.09	050/30S 025/35S 100/15S
MM	33.91	115.66	026/15–25W Nil
PW	33.93	115.41	096/15S
DB (NE)	33.58	115.63	017–032/10–18E
DB (SW)	33.58	115.70	010/15W 150/23W 297/33S plunge 265–292/17–19
CHB	33.64	115.60	vertical dike trending 045
CAA	33.51	115.55	125/30S
ZL	33.41	115.18	rhyolite plug
SC	33.44	115.16	007/26E 017/26E

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TABLE 2. PALEOMAGNETIC DATA FROM THE EASTERN TRANSVERSE RANGES

Locality	No. flows	No. samples	Mean* inclination ± ΔI	Mean* declination ± ΔD	Kappa	Alpha-95	R ± ΔR	F ± ΔF	Virtual geomagnetic pole Lat.	Virtual geomagnetic pole Long.	Potassium-argon date (Ma)
Pinto basin—clockwise declinations											
SW	8	42	62.6 ± 2.6	45.8 ± 5.6	45	3.3	47.2 ± 6.8	-10.1 ± 3.2	53.4N	56.0W	11.7 ± 4.7† 14.8 ± 1.3§
EA	5	18	60.8 ± 2.5	39.8 ± 5.1	585	3.2	41.2 ± 6.4	-8.3 ± 3.1	57.9N	51.7W	10.2 ± 2.0§
MW	7	30	38.4 ± 5.5	42.8 ± 8.4	15	7.0	44.2 ± 9.2	14.1 ± 5.8	50.5N	18.9W	··
PM	5	17	-49.6 ± 7.0	210.0 ± 10.8	16	9.1	31.4 ± 11.5	2.9 ± 7.2	64.5N	25.7W**	13.7 ± 1.6§
MM	2	11	24.1 ± 3.8	40.8 ± 4.1	71	5.4	42.2 ± 5.6	28.4 ± 4.2	47.3N	5.8W	··
Southern region											
PW††	8	52	56.1 ± 2.2	355.4 ± 3.9	50	2.8	-3.2 ± 5.5	-3.6 ± 2.9	85.4N	168.0W	7.8 ± 0.7§
DB (NE)††	9	54	34.1 ± 7.4	105.9 ± 9.0	32	9.3	107.0 ± 9.4	17.2 ± 7.9	2.2S	50.0W	··
DB (SW)††	6	36	37.2 ± 8.4	162.7 ± 10.6	42	10.5	164.1 ± 11.0	14.1 ± 8.8	33.2S	96.3W	··
CHB††,§§	1	3	48.4 ± 6.2	44.9 ± 9.3	226	8.2			51.7N	31.8W	··
CAA††,§§	1	6	-42.6 ± 7.3	250.7 ± 10.0	51	9.4			28.8N	37.6W**	19.0 ± 0.6***
ZL††,§§	1	8	-51.7 ± 2.7	80.0 ± 4.4	244	3.5			9.9N	172.8W**	24.5 ± 1.7†††
Region									Paleomagnetic pole		
(27 units)*			50.4 ± 5.1	39.7 ± 6.7	20	6.4	41.4 ± 7.7	2.1 ± 5.4	56.7N	30.7W	
(5 localities)§§§			47.3 ± 12.2	39.7 ± 18.3	24	15.9	41.1 ± 18.7	5.2 ± 12.3	55.8N	26.1W	

Note: late Miocene pole from Irving and Irving (1982) used as reference pole for North America (88N, 101E, $\alpha_{95} = 3$); determination of ±ΔI, ±ΔD, R ± ΔR, F ± ΔF from conventions of Beck (1980) as modified by Demarest (1983).
*Mean based on average of unit (flow) data.
†Plagioclase determination, San Diego State University.
§Whole-rock determination, San Diego State University.
**Northern hemisphere equivalent.
††Not included in regional pole determination.
§§Mean based on average of sample data.
***Data from Crowe and others (1979).
†††Sanidine determination, San Diego State University.
§§§Mean based on average of location data.

event to this time span. Given recent activity on faults of the Mojave Desert and eastern Transverse Ranges, however, we infer that the rotation of the ranges is continuing today (this is discussed more fully below). This would imply that PW is not part of the rotated block. Alternatively, the proximity of PW to both east-west sinistral and northwest-trending dextral faults (Fig. 2) suggests that opposing movements on these faults may have resulted in no net rotation at this locality.

MAGNETIZATION EXPERIMENTS

A detailed account of the experiments conducted on representative samples from each locality is given in Carter (1985). Stable characteristic directions are almost wholly associated with samples having median destructive a.f. demagnetization fields (MDF) greater than 10 mT. Additionally, some unstable samples exhibited abnormally high magnetizations which we attribute to isothermal remanent magnetization (IRM) due to lightning strikes (see McElhinny, 1973; Terres, 1984). Magnetic-hysteresis experiments show that those rocks exhibiting the softest and least stable demagnetization behavior are much nearer a multi-domain classification than are the others (criteria of Day and others, 1977). Acquisition of IRM and Curie point determinations indicate low-titanium magnetite as being the chief magnetic carrier in samples from most localities. These results are consistent with thermal demagnetization data indicating blocking temperatures rarely exceeding 580 °C. Exceptions are samples from the EA locality which in addition to exhibiting significant magnetization beyond 580 °C after demagnetization, still acquired IRM in magnetizing fields significantly greater (as much as 570 mT) than those expected for the saturation of magnetite (Terres, 1984). These observations suggest the presence of hematite. Although thermal demagnetization results from Smoketree Wash samples suggest the presence of hematite as well, IRM experiments indicate that it is not as significant as in the samples from the EA locality. For both of the localities, the directional data from the hematite component are concordant with the lower blocking temperature and a.f. results, thereby suggesting an early, deuteric alteration and the recording of an essentially primary magnetization.

DISCUSSION

Rotation Models

Powell (1981) suggested that the fault sets of the Mojave Desert and eastern Transverse Ranges are conjugate, resulting from north-northeast– to northeast-directed compression. Assuming an initial angle of 60° between the fault sets, Powell (1981) inferred that the originally north-northwest–trending faults of the Mojave region have rotated 30°–35° counterclockwise, whereas the originally east-northeast–trending faults of the eastern Transverse Ranges have rotated 30°–35° clockwise, resulting in the present angle of ~130° between them. In Garfunkel's (1974) model, which is linked to east-west sinistral shear due to basin-and-range extension, the late Cenozoic left-lateral faults of the eastern Transverse Ranges would take up strain via sinistral slip without rotating, whereas the originally north-trending, dextral faults of similar age in the Mojave Desert would experience 30°–40° of counterclockwise rotation.

Luyendyk and others (1980) interpreted the east-west–trending sinistral faults of the Transverse Ranges as originating as north-south–trending tension gashes within a northwest-trending, dextral shear couple resulting from Pacific and North American plate interactions. Alternatively, Luyendyk and others (1985a) suggested that the proto–Transverse Ranges originated as north-south–trending basin-and-range extensional features resulting from the subduction of the Farallon-Pacific spreading center. Because both models imply an 80°–90° clockwise rotation of the entire Transverse Ranges region, the smaller rotation suggested by paleomagnetic data for the eastern Transverse Ranges presents a conflict. This of course assumes that the rotation recorded by the upper Miocene basalts reflects the total Neogene rotation experienced by the region. Part of the very large clockwise declination anomaly seen in the Diligencia basin could be a regional rotation.

Geologic evidence suggesting a northeast-directed principal stress regime before Neogene time includes northeast-trending mineralization lineations and northwest-trending fold axes within the Pelona-Orocopia schist. These features are apparently related to late Mesozoic–

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early Cenozoic, northeastward movement along the Vincent–Chocolate Mountains thrust system which lies to the southwest of the Mojave Desert and eastern Transverse Ranges (Haxel and Dillon, 1978; Powell, 1981).

Since the middle Cenozoic, the region has apparently remained under a northeast-directed principal stress regime, as dextral shear was initiated between the Pacific and North American plates whereas sinistral shear was acting across the region as a result of basin-and-range extension (Garfunkel, 1974). Historic seismicity and evidence of Quaternary deformation have been recorded along faults of both the eastern Transverse Ranges and Mojave Desert regions, indicating synchronous activity and therefore supporting the idea that the faults are conjugate (Hope, 1966; Garfunkel, 1974; Fuis and others, 1977).

Many of the basalt fields in the eastern Transverse Ranges have been mapped as Quaternary and thought to be as young as Pleistocene (Hope, 1966). K-Ar dates which were determined in this study, however, indicate that they are late Miocene in age (Table 2, Fig. 2). This implies that the faulting which post-dates their emplacement is not necessarily restricted to the Quaternary as originally assumed. Small offsets recorded in Quaternary alluvial-fan deposits (Hope, 1966) and geomorphic observations based on Landsat Thematic Mapper imagery (R. Crippen, 1984, personal commun.) indicate that much of the displacement along major left-lateral faults of the eastern Transverse Ranges (that is, the Pinto Mountain and Blue Cut faults in particular) must have occurred prior to the Quaternary.

Amount of Rotation and Fault Slip

The net regional rotation can be computed by using an equation of Freund (1974) relating the observed left slip on the faults between the Pinto Mountain and Salton Creek faults (~50 km; Hope, 1966, 1969; Dibblee, 1968; Powell, 1981) to the width (north-south) of the entire region (70 km). This calculation predicts a 35°

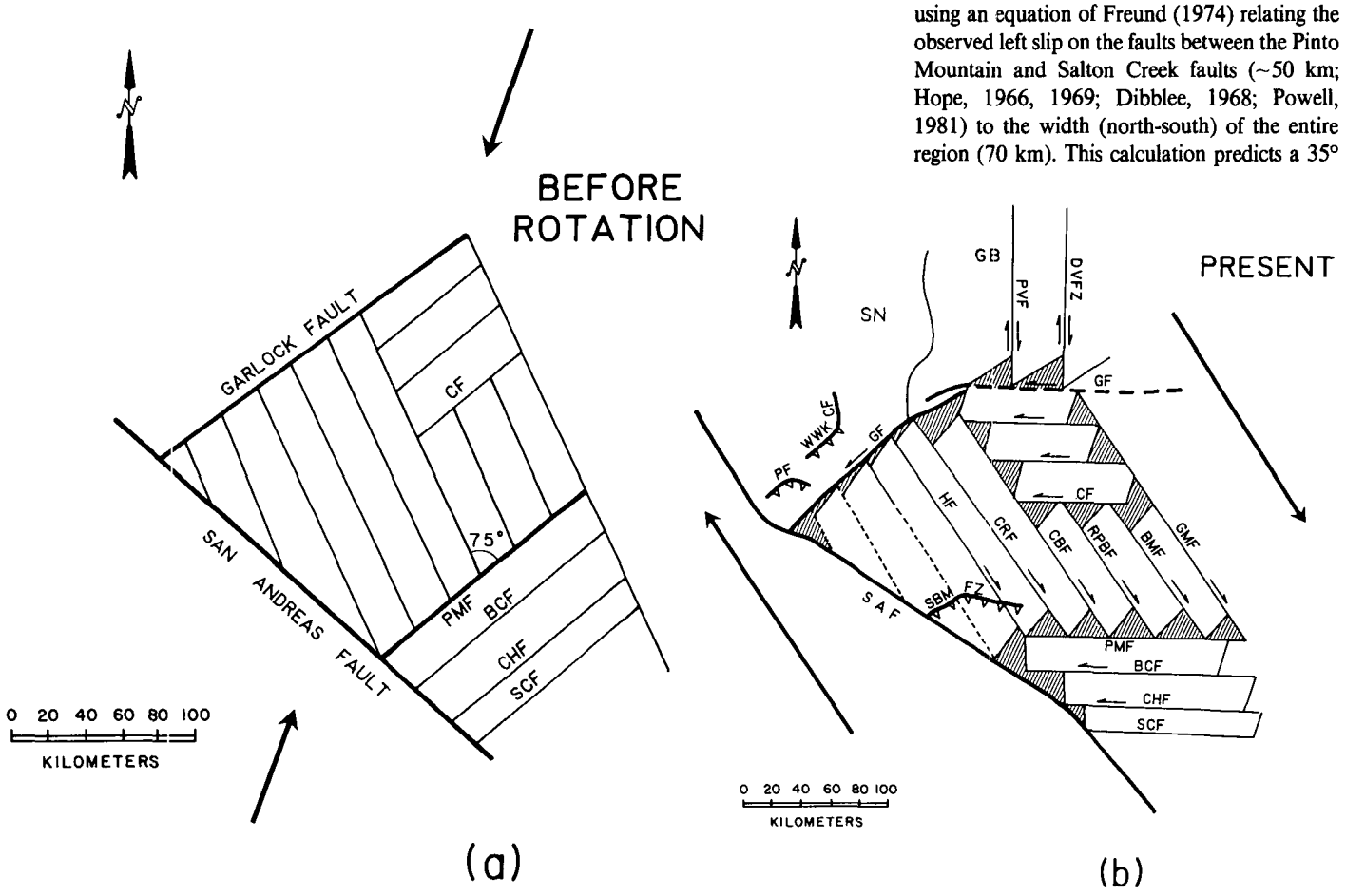


Figure 7. Rotational model for the eastern Transverse Ranges and Mojave Desert regions. (a) Late Miocene, pre-rotation configuration showing inferred conjugate fracture pattern resulting from a northeast-southwest maximum compressive stress. (b) Response of fracture system to northwest-directed dextral shear. This illustration is not meant to depict slip on individual faults; it is meant to illustrate the cumulative, regional shear necessary to comply with rotational constraints because smaller faults and elements of nonrigid behavior contribute to the regional shear. Similarly, triangular basins need not always form between the rotating blocks. Symbols: PMF = Pinto Mountain fault, BCF = Blue Cut fault, CHF = Chiriaco fault, SCF = Salton Creek fault, GMF = Granite Mountain fault, BMF = Bristol Mountains fault, RPBF = Rodman/Pisgah/Bullion fault system, CBF = Blackwater/Calico fault system, CRF = Camp Rock fault, HF = Helendale fault, SAF = San Andreas fault, SBMFZ = San Bernardino Mountains fault zone, PF = Pleito thrust fault, WWKCF = White Wolf–Kern Canyon fault, GF = Garlock fault, DVFZ = Death Valley fault zone, PVF = Panamint Valley fault, CF = Cady fault, SN = Sierra Nevada batholith, GB = Great Basin. (Names of faults from their associated references in the text and/or the California state sheets previously referenced.)

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clockwise rotation. This suggests that the 41° late Miocene rotation indicated by the paleomagnetic data probably reflects the total net rotation experienced by the eastern Transverse Ranges.

A reconstruction which restores slip on the sinistral faults of the eastern Transverse Ranges while back-rotating the region 40° counterclockwise indicates that the total distributed dextral shear across the Mojave Desert region must be on the order of 100 km to accommodate the rotation while remaining in contact with the eastern Transverse Ranges region (Fig. 7). This is in agreement with geologic evidence suggesting that cumulative slip across the region is on the order of 100 km (Garfunkel, 1974; Dibblee, 1961, 1971; Powell, 1981). These slip estimates also indicate that a larger net clockwise rotation of the eastern Transverse Ranges is unlikely. Dokka (1983) cited evidence that the cumulative slip on the major faults of the region is only between 26 and 38 km. The rotational constraints imposed from the results of this study, however, imply greater slip if these same faults accommodated most of the regional shear.

Additional Ramifications

In addition to the constraints on the amount of rotation and slip on the faults of the eastern Transverse Ranges, the reconstruction of Figure 7 also allows for a 15° counterclockwise rotation of the Mojave region. Although geometrical considerations led Garfunkel (1974) to suggest a greater rotation, paleomagnetic data from Burke and others (1982) and J. Morton and J. Hillhouse (unpub. data) suggest 15° ± 11° of net counterclockwise rotation since middle Miocene time. Additional counterclockwise rotation would necessarily require significantly greater slip on the dextral faults of the Mojave Desert.

A 15° counterclockwise rotation would place the initial angle between the fault sets of the Mojave Desert and eastern Transverse Ranges as being ~70° (Fig. 7a) and therefore very close to the angle expected between conjugate shears as suggested by brittle-failure theory and laboratory experiments (Paterson, 1978). The fact that the segment of the San Andreas fault bounding the region to the west is now oriented more northwest than its northern and southern segments may also be explained by such a counterclockwise rotation of the Mojave region (as originally suggested by Garfunkel, 1974).

Within this block-rotation scheme, triangular shaped gaps open if we assume perfectly rigid behavior of the blocks (Luyendyk and others, 1980). Isostatic gravity anomalies mapped by Roberts and others (1981) show gravity lows occurring just north and generally along the trend of the Pinto Mountain fault and in regions to the east and southeast of the proposed rotated region. These anomalies may be associated with basins filled with lower density sediments.

One last corollary to our model is that the Garlock fault which forms the northern boundary of the Mojave Desert region should also rotate clockwise (Fig. 7). The Garlock fault probably became active in the late Miocene (Carter, 1971; Burbank and Whistler, 1986), being contemporaneous with the faults of the eastern Transverse Ranges and Mojave Desert regions. Only the eastern segment of the Garlock north of the Cady Mountains region (predicted to be clockwise rotated by Luyendyk and others, 1980, and the prediction supported by preliminary studies of Luyendyk and others, 1985b), however, is inferred to have been rotated clockwise. For this rotation to occur, distributed dextral shear across the region immediately to the north (southwestern Great Basin) is also necessary and has probably been accommodated as dextral slip along north-northwest–trending faults, such as those of the Death Valley fault zone (Burchfiel and Stewart, 1966; Wright and Troxel, 1969).

To the west, the Garlock fault strikes more southwesterly; data from Miocene rocks in the Tehachapi Mountains immediately to the north (southeast of the White Wolf–Kern Canyon fault, Fig. 1) indicate no significant rotation since middle Miocene time (Kanter and McWilliams, 1982). Paleomagnetic results from Upper Cretaceous plutons to the southwest of this location indicate a 45° to 59° clockwise rotation since the Late Cretaceous (McWilliams and Li, 1985). Plescia and Calderone (1986) found an apparent 40° clockwise rotation for Oligocene flows nearby. They believed that it is probable that the rotation is post–Late Cretaceous but pre-dates 16 Ma.

The smaller (or less traceable) northwest-trending faults of the western portion of the Mojave Desert are probably not as significant as those to the east. It seems plausible that dextral shear strain over this region was taken up partially as crustal shortening along such features as the White Wolf and Plieto thrust systems to the north of the Garlock (Oakeshott, 1955; Crowell, 1964) and the fault system forming the northern rim of the San Bernardino Mountains to the south (Fig. 7b). The Sierra Nevada province to the north of this region may have acted as a buttress constraining dextral shear and, thus, the clockwise rotation of this segment of the Garlock.

SUMMARY AND CONCLUSIONS

The paleomagnetic data collected from the eastern Transverse Ranges region suggest that the region has undergone a net clockwise rotation of ~41° since late Miocene time. Results of age dating indicate that the basalt fields previously mapped as Quaternary (Hope, 1966) are actually late Miocene in age, implying that faulting in the region may have begun in late Miocene time. In addition, the clockwise rotation of the region constrains the deformation of the Mojave Desert to the north by (1) requiring that the cumulative slip on the northwest-trending dextral faults spanning the Mojave Desert be at least on the order of 80–100 km to accommodate the clockwise rotation and (2) suggesting that any counterclockwise rotation of the Mojave be constrained to be less than ~15° to stay within bounds of geologic evidence for maximum slip on the dextral faults. Conversely, these maximum slip estimates and those slips determined by mapping the faults of the eastern Transverse Ranges constrain the Neogene clockwise rotation to be ~40° on the basis of a geometric analysis.

It now seems that the amount of clockwise rotation of the entire Transverse Ranges region decreases systematically eastward, which is at variance with the model of Luyendyk and others (1980). Also, the rotation experienced by the eastern Transverse Ranges may have begun later than that experienced by the western Transverse Ranges. The eastern Transverse Ranges rotation has apparently occurred during the past 10 m.y.; the maximum age for the initiation of this deformation is constrained by the age of the youngest locality rotated, which is 10.2 ± 2.0 m.y. B.P. (locality EA, Table 1). The southern and eastern boundaries of the rotated region discussed in this study are still not clearly defined.

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